

1. Administrative & Study Identification

Official Study Title: A Phase III, Open-Label, Randomized Study of On-Demand TACE Combined With Atezolizumab Plus Bevacizumab (Atezo/Bev) or On-Demand TACE Alone in Patients With Untreated Hepatocellular Carcinoma **Brief Title:** A Study of TACE Combined With Atezolizumab Plus Bevacizumab or TACE Alone in Patients With Untreated Hepatocellular Carcinoma **Short Title/Acronym:** TALENTACE **Protocol Number:** ML42612 **NCT Number:** NCT04712643 **Trial Status:** Active, not currently recruiting (ACTIVE_NOT_RECRUITING) **Sponsor/Company:** Hoffmann-La Roche (Roche) **Investigational Product:** Atezolizumab (Tecentriq; ATC Code: L01FF05) + Bevacizumab (Avastin) + Transarterial Chemoembolization (TACE) **EMA Information:** [Atezolizumab EPAR Product Information](#) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of atezolizumab plus bevacizumab combined with on-demand TACE compared to on-demand TACE alone in participants with hepatocellular carcinoma (HCC) who are at high risk of poorer outcome following TACE treatment. **Secondary Objectives:** To evaluate investigator-assessed time to untreatable (unTACEable) progression (TTUP), time to progression (TTP), time to extrahepatic spread (EHS), Progression-Free Survival (PFS) by RECIST 1.1, Objective Response Rate (ORR), Duration of Response (DOR), and safety profiles.

2.2 Background & Rationale To address the gap in treatment for patients with systemically untreated, intermediate-to-high tumor burden, unresectable HCC who are at high risk of poorer outcome following TACE treatment. The IMbrave150 trial showed significantly improved OS with atezolizumab plus bevacizumab versus sorafenib. TALENTACE aims to demonstrate that combining this systemic immunotherapy with on-demand TACE can improve progression-free survival and overall survival compared to TACE alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (On-demand TACE alone) **Blinding:** Open-label **Randomization:** 1:1 **Trial Status:** Active but not currently recruiting

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 342 patients **Disease Area:** Hepatocellular Carcinoma (Liver Carcinoma) **Number of Sites:** Multicenter study across various sites (Predominantly in China, including Beijing, Shanghai, Guangzhou, Chengdu, etc.) **Estimated Study Start:** 12-Mar-2021 **Estimated Primary Completion:** 01-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Confirmed diagnosis of Hepatocellular Carcinoma (HCC) by histology/cytology or clinical criteria. 3. Unresectable disease, not a candidate for curative therapy. 4. Eligible for TACE treatment, with tumor max diameter + tumor number ≥ 6 . 5. No prior systemic therapy for HCC (especially immunotherapy) and no prior locoregional therapy to the target lesion(s). 6. At least one measurable untreated lesion. 7. ECOG Performance Status of 0-1. 8. Child-Pugh class A.

4.2 Exclusion Criteria 1. Evidence of Vp3/4 and hepatic vein tumor thrombus (HVTT). 2. Evidence of extrahepatic spread (EHS). 3. Being a candidate for curative treatments. 4. Any condition representing a contraindication to TACE as determined by the investigators. 5. Active or history of autoimmune disease or immune deficiency. 6. Untreated or incompletely treated esophageal and/or gastric varices with bleeding or high risk for bleeding. 7. A prior bleeding event due to esophageal and/or gastric varices within 6 months prior to initiation of study treatment. 8. Evidence of bleeding diathesis or significant coagulopathy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab (Tecentriq, 1200 mg) and bevacizumab (15 mg/kg) administered on Day 1 of each 21-day cycle. TACE is performed on clinical demand. **Route of Administration:** Intravenous (IV) for atezolizumab and bevacizumab; Transarterial for TACE. **Duration of Treatment:** Until untreatable progression, TACE failure/refractoriness, unacceptable toxicity, or loss of clinical benefit.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for underlying liver disease and symptom management. **Prohibited:** Other concurrent systemic anti-cancer therapies and live-attenuated vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent,

Medical History, Tumor Baseline Imaging | | **Baseline** | Day 0 | Randomization, First Dose (Atezo/Bev + TACE or TACE alone) | | **Interim** | Day 1 of each 21-day Cycle | IV Infusions, Safety Labs, Efficacy Assessment (Imaging), AE tracking | | **End of Study** | Follow-up/Study Completion | Final Vital Signs, Survival Follow-up, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. TACE Progression-Free Survival (TACE PFS) as Determined by Investigator:** Defined as the time from randomization to untreatable progression, TACE failure/refractoriness, or any cause of death, whichever occurs first.
- 2. Overall Survival (OS):** Defined as the time from randomization to death from any cause after enrollment.

7.2 Secondary & Exploratory Outcomes

- 1. Time to Untreatable (unTACEable) Progression (TTUP):** As determined by Investigator.
- 2. Time to Progression (TTP):** As determined by Investigator.
- 3. Time to Extrahepatic Spread (EHS):** As determined by Investigator.
- 4. Progression-free survival (PFS)** according to RECIST v1.1.
- 5. Objective response rate (ORR) and Duration of Response (DOR).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during study visits, graded by CTCAE. Treatment-related AEs associated with TACE, atezolizumab, and bevacizumab are tracked. **Serious Adverse Events (SAEs):** Prompt reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to review safety and efficacy data across intervals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population (\$N=342\$; 171 per arm) for primary efficacy outcomes. Safety Set for all participants who received at least one dose of study treatment. **Sample Size Justification:** Powered to detect a statistically significant improvement in TACE-PFS and OS for the combination arm versus TACE alone. **Handling of Missing Data:** Standard censoring rules for time-to-event endpoints (Kaplan-Meier methods).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring to ensure protocol compliance and data integrity. **Audit Trail:** Secure eCRF entry, tracking of radiographic reads by independent central review, and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: ML42612 **Registry Number:** NCT04712643 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** TALENTACE **Official Regulatory Title:** A Phase III, Open-Label, Randomized Study of On-Demand TACE Combined With Atezolizumab Plus Bevacizumab (Atezo/Bev) or On-Demand TACE Alone in Patients With Untreated Hepatocellular Carcinoma

12. Clinical Framework (HCC Specific)

Primary Condition: Hepatocellular Carcinoma (HCC) **Diagnosis Threshold:** Unresectable HCC, eligible for TACE, without extrahepatic spread (EHS) or macrovascular invasion (HVTT/Vp3/4) **Medical Methodology:** Locoregional therapy (TACE) combined with systemic immunotherapy and VEGF inhibition **Optimization Target:** Improvement in TACE Progression-Free Survival (TACE-PFS) and Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: On-demand TACE combined with Atezolizumab and Bevacizumab every 21-day cycle **Education Protocol (HCP):** Site initiation and continuous training on RECIST v1.1 and unTACEable progression criteria **Education Protocol (Patient):** Informed consent processes, scheduling compliance, and recognition of immune-mediated or bleeding AEs **Quality Audit Mechanism:** Blinded Independent Review Committee (IRC) for radiographic endpoints and AE verification

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until study completion (Est. 2029) or death **Onsite Visit Cadence:** Every 21 days (Day 1 of each cycle) for systemic therapy administration **Remote Monitoring Frequency:** Regular survival tracking post-treatment discontinuation **Checklist Review Interval:** Routine tumor imaging assessments according to protocol (e.g., every 6-9 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					